

数学作业纸

科目_____

班级:

姓名:

编号:

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$$\psi_{116} \tau_{max} = \frac{T}{n_c} = \frac{M_e}{W_e} \leq [\tau]$$

$$M_e \leq W_e [\tau] = \frac{\pi D^3}{16} \left[1 - \left(\frac{d}{D} \right)^4 \right] \cdot [\tau]$$

$$= \frac{3.14}{16} \times 50^3 \times 10^9 \times \left[1 - \left(\frac{38}{50} \right)^4 \right] \times 70 \times 10^6$$

$$\approx 11448 \text{ N}\cdot\text{m} \approx 11.45 \text{ kN}\cdot\text{m}$$

$$\varphi_{AB} = \frac{T a}{G I_{p1}} \quad \varphi_{BC} = \frac{T b}{G I_{p2}}$$

$$\varphi_{AB} = \varphi_{BC}$$

$$\frac{T a}{G I_{p1}} = \frac{T b}{G I_{p2}}$$

$$\frac{a}{b} = \frac{I_{p1}}{I_{p2}} = \frac{\frac{\pi D^4}{32} \left[1 - \left(\frac{d}{D} \right)^4 \right]}{\frac{\pi d^4}{32} \left[1 - \left(\frac{d_1}{d} \right)^4 \right]} = \frac{1 - \left(\frac{d}{D} \right)^4}{1 - \left(\frac{d_1}{d} \right)^4} = \frac{1 - \left(\frac{25}{50} \right)^4}{1 - \left(\frac{38}{50} \right)^4} \approx 1.40$$

$$a/b = 1.4$$

$$\begin{cases} a = 297.5 \text{ mm} \\ b = 212.5 \text{ mm} \end{cases}$$